NeuraQ — measuring and mitigating barren plateaus on real quantum hardware

NeuraQ · Qiskit Fall Fest 2026

Problem

Variational quantum neural networks (QNNs) are trained by estimating the gradient of a cost function with respect to the circuit parameters. For deep, randomly initialised circuits that gradient vanishes **exponentially** with the number of qubits: the barren plateau. This is not merely "small gradients". At a plateau a fixed number of gradient measurements carries an exponentially vanishing amount of information about the cost, so adding shots or iterations buys nothing and training stalls. The failure mode is precisely the regime people want: deep, expressive circuits trained on a global cost.

The goal of NeuraQ is a **measured, error-barred** comparison of what actually controls the plateau — the locality of the cost function and the standard mitigations — validated not only in an ideal simulator but also under device noise and on a real IBM Quantum processor. Success means a reproducible, honest number, not an advantage claim.

Method

Simulator. The core is a batched NumPy statevector simulator (little-endian basis, matching Qiskit). Gradients use reverse-mode **adjoint differentiation** and are validated three ways: against parameter-shift, against finite differences, and against Qiskit Aer (E0). The ansatz is a single-axis layered hardware-efficient ansatz, $R_1\ E\ R_2\ E\ \dots\ E\ R_M$, with a CZ chain, the standard structure used to exhibit plateaus.

Protocol. For the target $|0\dots0\rangle$ we use two costs on the same circuit:

- global (faithful): $C_G = 1 - P(0\dots0)$, an observable acting on all qubits;
- local: $C_L = 1 - (1/n)\ \Sigma_i\ P_i(0)$, a sum of single-qubit observables.

We estimate R random initialisations and report $\mathrm{Var}(\partial C/\partial\theta)$ over the middle layer, with **bootstrap 95% confidence intervals**, and fit the decay as a log-linear slope per qubit. C_L and C_G obey $C_L \le C_G \le n \cdot C_L$, which turns the trainable local cost into a bound on the faithful global one (E19).

Mitigations evaluated on equal footing. Local cost (E2a); identity-block and **perturbed-identity** initialisation (E2b, E20); layerwise training (E2b); support scheduling / Cost-Support Annealing (E13); and a **bounded local→global λ-mix **schedule (E19). A** trainability screen** packages the same measurement as a pre-flight verdict (E16, E18).

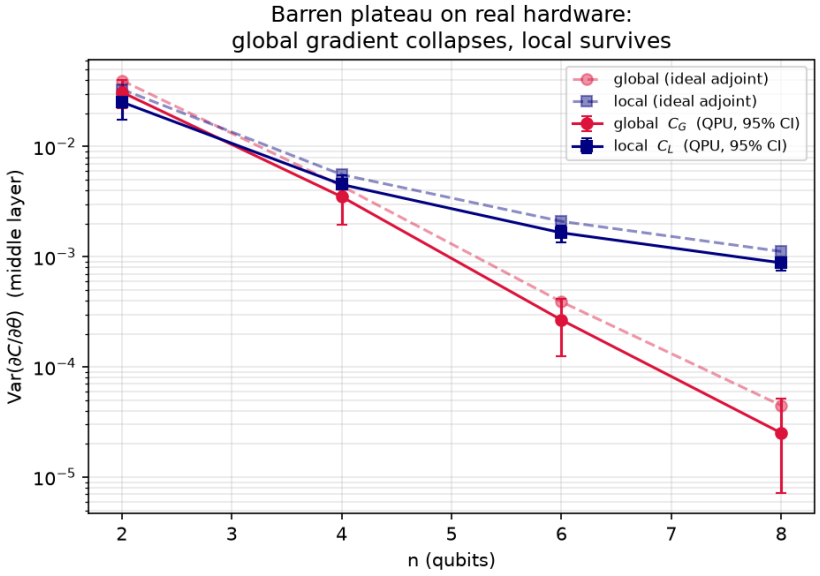
Beyond the ideal simulator. We repeat the measurement with a **FakeBrisbane device-noise model** in Aer, and once on real hardware: IBM Quantum ibm_fez, $n \in \{2,4,6,8\}$, depth $M = 4$, $R = 48$ inits, 512 shots (jobs db76ns2raf…, db76pn8r…, db76qfqr…, db76rl6r…, all DONE, 2026-10-13). All simulator results are exact; the QPU variances include shot noise and are reported as such. Exact diagonalisation and an L-BFGS-B oracle give classical references for the energy tasks.

Results

The plateau, and the locality lever, on real hardware. The hero figure (figures/main_result.png) shows gradient variance versus width for the global and local costs. The global-cost variance collapses by roughly 4^-n; the local-cost variance decays far more slowly.

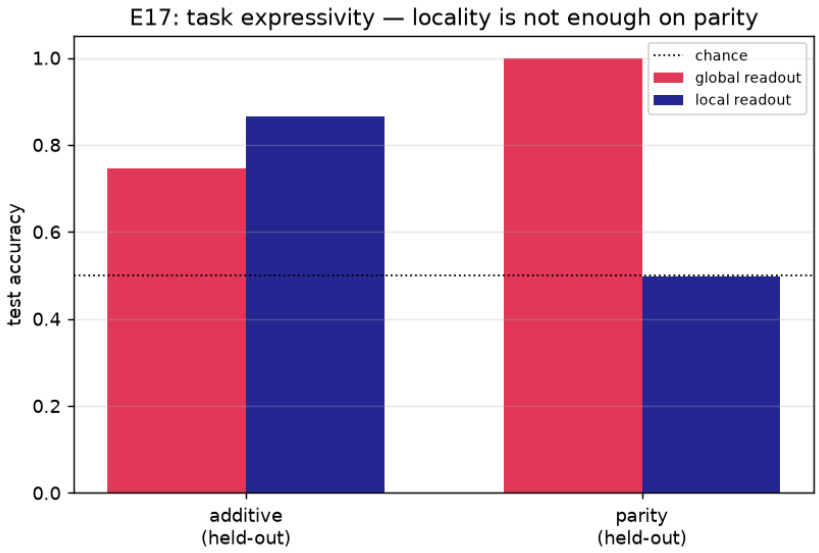
n	global Var (sim)	global Var (noise)	global Var (QPU)	local Var (QPU)	local / global (QPU)
2	4.0e-2	3.4e-2	3.1e-2	2.5e-2	1.2
4	4.4e-3	3.3e-3	3.5e-3	4.5e-3	0.8
6	3.9e-4	2.7e-4	2.7e-4	1.7e-3	6.2
8	4.7e-5	3.0e-5	2.5e-5	8.8e-4	35

The fitted slopes are: ideal simulation global −1.16, local −0.55 per qubit; device-noise global −1.21, local −0.56; real hardware global −1.20, local −0.55. The full-window simulator sweep (E2a, R = 512, n = 2…12) gives global −1.27 (95% CI [−1.34, −1.22]) and local −0.45 ([−0.449, −0.443]). At n = 8 the local/global gap is ≈25× in the ideal simulation, ≈31× under device noise, and ≈35× on the QPU. Noise shrinks both variances (the signal never grows) but leaves the slopes intact: **the hardware data is consistent with the noise model, and the locality gap is at least as large as simulated, with overlapping confidence intervals.**



Barren plateau on real hardware

Locality is not expressivity. A subtler and cleaner result (E17): on the additive task (E6) the local readout wins (0.87 vs 0.75 held-out accuracy), but on a **parity** task the local readout is provably unable to represent the label — it scores chance (0.50) — while the global readout solves it exactly (1.00, held-out). So "use a local cost" is not a free win: the trainable cost is the one that can express the target, which is why the harness must expose both.

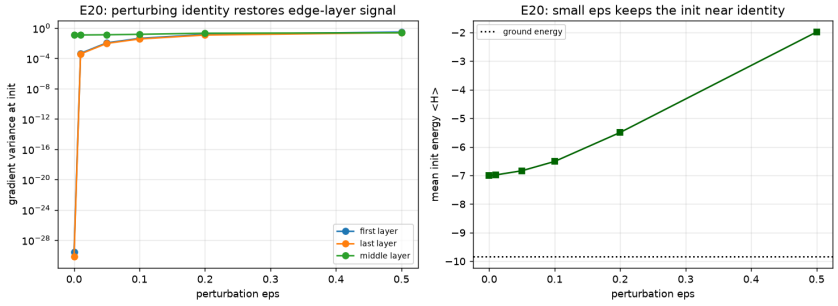


Task expressivity

Support radius (E11). Generalising beyond binary local/global, gradient variance decays approximately exponentially in the observable's support: adding one qubit of support multiplies the variance by ≈ 1/2^(κ) with κ ≈ 1.6–2.0 × ln 2 for n = 6…12. This predicts noise tolerance directly from an observable's support.

A bounded schedule (E19). The λ-mix cost (1−λ)C_L + λC_G was tested with λ ramped 0→1. The normalisation bound C_L ≤ C_G ≤ n·C_L holds empirically (max violation 0; max ratio C_G/C_L = 2.8 ≤ n). Under both exact and Adam-normalised shot gradients all arms reach a low global cost; the blend is never worse than the global cost and always starts on the trainable surface.

Perturbed identity (E20). The exact identity block leaves the first and last layers with essentially no gradient signal at initialisation (first-layer variance ≈ 2.6e-30). Perturbing every angle by U(−ε, ε) restores it (4.6e-4 at ε = 0.01, 1.5e-1 at ε = 0.2) while keeping the initial energy near the identity value.



Perturbed identity init

Screening (E16/E18). The screen separates widths cleanly (slope −1.27 per qubit; recommends a local cost with a ≈37× variance improvement), but its verdict is a gradient-variance alarm, not an optimiser-failure predictor at these widths (see limits).

Honest limits

- **What is simulated.** Exact noiseless NumPy statevector; Qiskit Aer noiseless; Aer with a **FakeBrisbane** device noise model — not ibm_fez specifically; and one real QPU run at small scale.

- **No advantage claim.** NeuraQ characterises trainability. It does not claim quantum advantage, speed-up, or a better classifier.
- **The plateau does not always block optimisers.** With Adam's per-parameter normalisation the global-cost arm can still reach a low cost at n = 8 (E3), and even under shot noise every configuration in the n ≤ 10 screen validation reached a low cost (E18). The screen's "barren" verdicts were **false negatives** there (TN = 0, FN = 3). The plateau is real and measurable as vanishing gradient variance, but at these widths it slows training rather than stopping it.
- **Relative, not asymptotic.** The local-cost slope (≈ −0.55 per qubit) is still exponential over n = 2…8; the claim is that it decays far more slowly than the global one, not that it is polynomial.
- **Weak results kept.** Cost-Support Annealing (E13) is partial: the linear schedule helps, the staged and variance-gated variants do not beat the plain global cost. The E11 averaged cost carries a 1/W normalisation that produces a mild non-monotonic shoulder at r = 1,2.
- **Classical baseline.** Energy tasks are checked against exact diagonalisation and an L-BFGS-B oracle, and the classifier against a **held-out** split; a matched-parameter-count classical MLP is the top remaining item and is not yet included.
- **Thresholds.** The screen's verdict thresholds were calibrated on the same sweeps it is judged against (E18).
- **Shot noise.** The E15 hardware variances include shot noise; the backend name, date and job IDs are recorded in the README.

AI-use note

Coding agents (opencode CLI) gave **mild assistance only** — scaffolding, boilerplate, docstrings, and running the test suite. One bug the team flagged and had fixed: the classifier's MSE gradient had the wrong sign (-2(y - p) instead of mean(-2(y - base) 0.5 (C(theta + pi/2) - C(theta - pi/2)))), which had flipped an earlier "local cost worse" conclusion; it was caught by the finite-difference check (agreement ~2e-10). Full note in AI_USE_NOTE.md.